

TrueGrain Deck™ Construction

RENOLIT laminate. Patwin cellular PVC. American-extruded.

The TrueGrain deck board is a precision-laminated, five-layer PMMA film stack engineered by RENOLIT, permanently bonded to a Patwin-extruded cellular PVC profile with a proprietary ASA capstock. Each layer of the film performs a distinct functional role from abrasion resistance to chemical adhesion. The substrate beneath it is American-extruded under one roof in Linden, New Jersey. This data sheet documents both halves of the assembly for architects, specifiers, dealers, and distributor partners.

● ENGINEERED BY RENOLIT

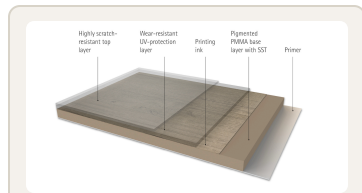


FIGURE 1.
TRUEGRAIN LAMINATE
CROSS-SECTION. LAYERS
SHOWN IN TRUE STACKING
ORDER, TOP TO BOTTOM.

PRODUCT	TrueGrain Deck™ capped cellular PVC
SURFACE	RENOLIT exterior PMMA laminate
FILM CHEMISTRY	PMMA (acrylic) with SST
BONDING	Primer-assisted permanent lamination
ORIGIN	Laminated and extruded in Linden, NJ, USA
WARRANTY	25-year limited residential

THE BOARD BENEATH THE FILM EXTRUDED BY PATWIN PLASTICS

- CAPSTOCK** Patwin proprietary ASA, formulated and co-extruded in-house. Caps all four sides of the profile.
- CORE** Cellular PVC with 30% in-house regrind. Closed-loop American material, single facility.
- PROCESS** PVC compound mixed in-house. Dies designed and machined in-house. Cellular PVC decking under one roof since 2001.

01 Highly scratch-resistant top layer

SURFACE DURABILITY

Clear, high-hardness exterior surface film engineered to resist abrasion from foot traffic, patio furniture, pet claws, sand, and grit. Defines the wear face of the board and protects the printed decor against everyday mechanical contact in residential service.

02 Wear-resistant UV-protection layer

WEATHERING & COLOR HOLD

Dedicated UV stabilizer film formulated to absorb solar radiation before it can reach the print or the polymer substrate. Preserves color fidelity on the decorative wood-grain and shields the underlying chemistry from photo-oxidative degradation.

03 Printing ink

PHOTOREALISTIC DECOR

High-resolution rotogravure decor print laid down between the protective films above and the pigmented PMMA base below. Reproduces variegated grain, pore structure, and tonal depth of premium hardwoods with edge-to-edge registration on every board.

04 Pigmented PMMA base layer with SST

HEAT REFLECTION & FILM BACKBONE

Acrylic (PMMA) base layer carrying RENOLIT's Solar Shield Technology (SST). SST reflects the near-infrared portion of the solar spectrum so the board runs measurably cooler under direct sun, while the PMMA chemistry provides the dimensional stability and weatherability specified for exterior cladding.

05 Primer

SUBSTRATE ADHESION

Engineered tie-coat applied to the underside of the film stack. Forms the chemical bond that locks the laminate to the capped cellular PVC substrate, producing a permanent lamination rather than a surface-applied coating.

RENOLIT engineers the film. Patwin Plastics extrudes the board. Patwin laminates the RENOLIT TrueGrain PMMA foil to its own proprietary ASA-capped cellular PVC substrate in Linden, New Jersey, marrying European film science to an American-extruded profile. The result is the twenty-five-year board specified by American Pro Building Products.

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